

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures Assessment Tool 2 – Debugging & Optimisation Task 100
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	
Weightage:	35% of CIA	Total Marks:	
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	Ragul.m	Reg. No.:	192524066 24/07/2026
Section / Batch:	Slot A	Date:	

Code of Conduct

I RAGUL.M (192524066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: RAGUL.M

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 - 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edgecase errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors

Code Quality & Readability	20	Clean, modular, well documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

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10/10/2024

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Identifying Symbols 5 marks
- Q6 Stack - Identifying Symbols 5 marks
- Q7 Stack 5 marks
- Q8 Stack 5 marks
- Q9 Stack 5 marks
- Q10 Stack 5 marks

Q1 Binary Search 5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int N, int key) {
    int low = 0, high = N - 1;
    while (low < high) {
        int mid = (low + high) / 2;
        if (arr[mid] == key)
            return mid;
        else if (arr[mid] > key)
            low = mid;
        else
            high = mid;
    }
    return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <iostream>
using namespace std;

int binarySearch(int arr[], int N, int key) {
    int low = 0, high = N - 1;
    while (low <= high) {
        int mid = (low + high) / 2;
        if (arr[mid] == key)
            return mid;
        else if (arr[mid] > key)
            high = mid - 1;
        else
            low = mid + 1;
    }
    return -1;
}

int main() {
    int arr[] = {5, 10, 15, 20, 25, 30, 35, 40};
    int N = sizeof(arr) / sizeof(arr[0]);
    int key = 20;
    int result = binarySearch(arr, N, key);
    if (result != -1)
        cout << "Element found at index: " << result << endl;
    else
        cout << "Element not found" << endl;
    return 0;
}
```

Submitted

Test Input

```
Enter Search Element: 20
```

Output

```
Element found at index: 4
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
Back

QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Identifying Symbols 5 marks
- Q6 Stack - Identifying Symbols 5 marks
- Q7 Stack 5 marks
- Q8 Stack 5 marks
- Q9 Stack 5 marks
- Q10 Stack 5 marks

Q2 Stack 5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
    if (top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <iostream>
using namespace std;

const int MAX = 5;
int stack[MAX];
int top = -1;

int pop() {
    if (top == -1) {
        printf("Underflow\n");
        return -1;
    }
    return stack[top--];
}

void push(int x) {
    if (top == MAX - 1) {
        printf("Overflow\n");
        return;
    }
    stack[++top] = x;
}

int main() {
    printf("Test1: pop() = %d\n", pop());
    push(10);
    printf("Test2: pop() = %d\n", pop());
    push(20);
    printf("Test3: pop() = %d\n", pop());
    push(30);
    printf("Test4: pop() = %d\n", pop());
    return 0;
}
```

Submitted

Test Input

```
stdin: Input Captured
```

Output

```
Test1: pop() = -1
Test2: pop() = 10
Test3: pop() = 20
Test4: pop() = 30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
Back

QUESTIONS

Q1
Stack Search
3 marks

Q2
Stack
3 marks

Q3
Stack
3 marks

Q4
Stack
3 marks

Q5
Stack - Returning Symbol
3 marks

Q6
Single linked list
3 marks

Q7
Stack
3 marks

Q8
Linked List
3 marks

Q9
Linked List
3 marks

Q10
Stack
3 marks

10/10 answered

Q2 Stack
3 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
int stack[5];
int top;
void init() {
    top = -1;
}
int main() {
    init();
    printf("top = %d\n", top);
    return 0;
}
```

Test Input

Output

```
top = -1
```

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Submitted

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
Back

QUESTIONS

Q1
Stack Search
3 marks

Q2
Stack
3 marks

Q3
Stack
3 marks

Q4
Stack
3 marks

Q5
Stack - Returning Symbol
3 marks

Q6
Single linked list
3 marks

Q7
Stack
3 marks

Q8
Linked List
3 marks

Q9
Linked List
3 marks

Q10
Stack
3 marks

10/10 answered

Q2 Stack
3 marks

Identify the bugs in the following snippet.

```
int top = 0;
void init(stack) {
    printf("top = %d\n", top);
}
void init() {
    top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#define MAX 100
char stack[MAX];
int top = -1;
void init(stack) {
    printf("top = %d\n", top);
}
void init() {
    top = -1;
}
int main() {
    if (top < 0) printf("top = %d\n", top);
    printf("top = %d\n", top);
    return 0;
}
void push(int x) {
    if (top < MAX) {
        printf("pushing %d\n", x);
        stack[top] = x;
        top++;
    } else {
        printf("stack is full\n");
    }
}
void pop() {
    if (top > -1) {
        printf("popping %d\n", stack[top]);
        top--;
    } else {
        printf("stack is empty\n");
    }
}
void display() {
    printf("Stack elements: ");
    for (int i = 0; i < top; i++) {
        printf("%d ", stack[i]);
    }
    printf("\n");
}
int main() {
    init();
    push(10);
    push(20);
    display();
    pop();
    display();
    return 0;
}
```

Test Input

Output

```
top = 0
pushing 10
pushing 20
Stack elements: 10 20
popping 20
Stack elements: 10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1
Binary Search
5 marks
- Q2
Stack
5 marks
- Q3
Stack
5 marks
- Q4
Stack
5 marks
- Q5
Stack - Balancing Symbols
5 marks
- Q6
Single Linked List
5 marks
- Q7
Stack
5 marks
- Q8
Linked List
5 marks
- Q9
Linked List
5 marks
- Q10
Queue
5 marks

10/10 answered

Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.

```
if(ch == '{') && stack[top] == '{') {  
    ch == '}' && stack[top] == '}' {  
        pop();  
    } else {  
        return 0;  
    }
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <iostream.h>  
#include <stack.h>  
using namespace std;  
int main() {  
    int n = 0;  
    char ch;  
    stack<char> s;  
    while (n < 10) {  
        ch = getchar();  
        if (ch == '{') s.push(ch);  
        else if (ch == '}') {  
            if (s.empty()) return 0;  
            char top = s.top();  
            if (top == '{') s.pop();  
            else return 0;  
        }  
        n++;  
    }  
    return 1;  
}
```

Test Input

stdin input (optional)

Output

```
Test 1: "({{{{}}})" -> Balanced  
Test 2: "({{{{}}})" -> Not  
Balanced  
Test 3: "({{{{}}})" -> Balanced  
Test 4: "({{{{}}})" -> Balanced
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

- Q1
Binary Search
5 marks
- Q2
Stack
5 marks
- Q3
Stack
5 marks
- Q4
Stack
5 marks
- Q5
Stack - Balancing Symbols
5 marks
- Q6
Single Linked List
5 marks
- Q7
Stack
5 marks
- Q8
Linked List
5 marks
- Q9
Linked List
5 marks
- Q10
Queue
5 marks

10/10 answered

Singly Linked List

5 marks

What is the issue?

```
temp = head  
while (temp->next != NULL) {  
    print(temp->data);  
    temp = temp->next;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <iostream.h>  
#include <conio.h>  
using namespace std;  
struct Node {  
    int data;  
    struct Node *next;  
};  
Node *head = NULL;  
Node *temp = NULL;  
Node *newNode(int data) {  
    Node *newNode = new Node;  
    newNode->data = data;  
    newNode->next = NULL;  
    return newNode;  
}  
Node *buildListFromArray(int arr[], int n) {  
    Node *head = NULL;  
    Node *temp = NULL;  
    for (int i = 0; i < n; i++) {  
        temp = newNode(arr[i]);  
        if (head == NULL) head = temp;  
        else head->next = temp;  
    }  
    return head;  
}  
void display(Node *head) {  
    while (head != NULL) {  
        cout << head->data << " ";  
        head = head->next;  
    }  
}
```

Test Input

stdin input (optional)

Output

10 20 30 40

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← BACK

QUESTIONS

- Q1
Binary Search
5 marks
- Q2
Stack
5 marks
- Q3
Stack
5 marks
- Q4
Stack
5 marks
- Q5
Stack - Balancing Symbols
5 marks
- Q6
Singly Linked List
5 marks
- Q7
Stack
5 marks
- Q8
Linked List
5 marks
- Q9
Linked List
5 marks
- Q10
Stack
5 marks

10/10 answered

Q7 Stack

5 MARKS

Point out the error.
POP(stack);
if top == 0:
print("Underflow")
else:
return stack[top-1]

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#define MAX 5
int top=-1;
int stack[MAX];

void push(int x) {
    if (top == MAX-1) {
        printf("Stack Overflow\n");
        return;
    }
    stack[++top] = x;
    printf("Pushed: %d\n", x);
}

void pop() {
    if (top < 0) {
        printf("Stack Underflow\n");
        return;
    }
    printf("Popped: %d\n", stack[top]);
    top--;
}

int main() {
    int x;
    printf("Enter a number to push: ");
    scanf("%d", &x);
    push(x);
    printf("Enter a number to pop: ");
    scanf("%d", &x);
    pop();
    return 0;
}
```

Submitted

Test Input

stdin input (optional):

Output

```
Test 1: POP on empty stack
Underflow
Test 2: Push 10, 20, 30 then
pop
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← BACK

QUESTIONS

- Q1
Binary Search
5 marks
- Q2
Stack
5 marks
- Q3
Stack
5 marks
- Q4
Stack
5 marks
- Q5
Stack - Balancing Symbols
5 marks
- Q6
Singly Linked List
5 marks
- Q7
Stack
5 marks
- Q8
Linked List
5 marks
- Q9
Linked List
5 marks
- Q10
Stack
5 marks

10/10 answered

Q8 Linked List

5 MARKS

Identify the errors in the following code
while current != NULL:
currentnext = prev
prev = current
current = current.next

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>
typedef struct Node {
    int data;
    struct Node* next;
} Node;

Node* createNode(int x) {
    Node* n = (Node*)malloc(sizeof(Node));
    n->data = x;
    n->next = NULL;
    return n;
}

void append(Node** head, int x) {
    if (*head == NULL) {
        *head = createNode(x);
        return;
    }
    Node* cur = *head;
    while (cur->next != NULL) cur = cur->next;
    cur->next = createNode(x);
}

void printList(Node* head) {
    while (head != NULL) {
        printf("%d ", head->data);
        head = head->next;
    }
    printf("\n");
}

Node* reverseList(Node* head) {
    Node* prev = NULL;
    Node* curr = head;
    while (curr != NULL) {
        Node* next = curr->next;
        curr->next = prev;
        prev = curr;
        curr = next;
    }
    return prev;
}

int main() {
    Node* head = NULL;
    append(&head, 1);
    append(&head, 2);
    append(&head, 3);
    append(&head, 4);
    printf("Original List: ");
    printList(head);
    head = reverseList(head);
    printf("Reversed List: ");
    printList(head);
    return 0;
}
```

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Test Input

stdin input (optional):

Output

```
Original List: 1 2 3 4
Reversed List: 4 3 2 1
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Monitoring System
5 marks

Q6
Single Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Q6: Linked List
5 marks

What is the error in the below code?
new.next = head
head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

#include <stdio.h>
#include <stdlib.h>
typedef struct Node {
    int data;
    struct Node *next;
} Node;

Node* createNode(int val) {
    Node* n = (Node*)malloc(sizeof(Node));
    n->data = val;
    n->next = NULL;
    return n;
}

Node* insertBeginning(Node* head, int val) {
    Node* newNode = createNode(val);
    newNode->next = head;
    head = newNode;
    return head;
}

void printList(Node* head) {
    Node* temp = head;
    while(temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    printf("\n");
}

int main() {
    Node* head = NULL;
    head = insertBeginning(head, 10);
    head = insertBeginning(head, 20);
    head = insertBeginning(head, 30);
    printList(head);
    head = insertBeginning(head, 40);
    printList(head);
    return 0;
}

```

Test Input
stdin input (optional)

Output
Before insertion: 10 20 30
After insertion: 40 10 20 30

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DS PROGRAMMING

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Monitoring System
5 marks

Q6
Single Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Q8: Queue
5 marks

What is missing in the below pseudocode?
DEQUEUE(Q)
if front == -1
print("Underflow")
else
return Q[front++]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

#include <stdio.h>
#define MAX 5
int queue[MAX];
int front = -1, rear = -1;
int isEmpty() {
    return (front == -1);
}

void enqueue(int x) {
    if ((front == 0 && rear == MAX - 1) || (rear + 1 == front)) {
        printf("Overflow\n");
        return;
    }
    if (front == -1) front = 0;
    rear = (rear + 1) % MAX;
    queue[rear] = x;
}

int dequeue() {
    if (isEmpty()) {
        printf("Underflow\n");
        return -1;
    }
    int x = queue[front];
    if (front == rear) {
        front = rear = -1;
    } else {
        front = (front + 1) % MAX;
    }
    return x;
}

int main() {
    printf("Enqueue: 10, 20, 30\n");
    enqueue(10);
    enqueue(20);
    enqueue(30);
    printf("Dequeue: 40\n");
    dequeue();
    printf("Dequeue: 30\n");
    dequeue();
    printf("Dequeue: 20\n");
    dequeue();
    printf("Dequeue: 10\n");
    dequeue();
    printf("Dequeue: 0 (should underflow): 40\n");
    dequeue();
    return 0;
}

```

Test Input
stdin input (optional)

Output
Enqueue: 10, 20, 30
Dequeue: 40
Dequeue: 30
Dequeue: 20
Dequeue: 10
Dequeue: 0

Pending manual review

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INDUJEEVARA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1 Stack 3 marks

Q2 Stack Overflow 3 marks

Q3 Stack 3 marks

Q4 Stack 3 marks

Q5 Variable Uninitialized var 3 marks

Q6 Queue 3 marks

Q7 Queue - Dequeue 3 marks

Q8 Binary Search 3 marks

Q9 Vector Queue 3 marks

Q10 Stack - Push Pop Equivalent 3 marks

10/10 answered

Q1 Stack 3 marks

Point out the error?
PEEK(stack);
return stack[top+1];

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <iostream.h>
2 #define MAX 100
3 int stack[MAX];
4 int top = -1;
5 void push(int value) {
6     if (top == MAX - 1) {
7         printf("Stack Overflow\n");
8         return;
9     }
10    stack[++top] = value;
11 }
12 int pop() {
13     if (top == -1) {
14         printf("Stack Underflow\n");
15         return -1;
16     }
17     return stack[top--];
18 }
19 int peek() {
20     if (top == -1) {
21         printf("Stack is empty\n");
22         return -1;
23     }
24     return stack[top];
25 }
26 int main() {
27     push(10);
28     push(20);
29     push(30);
30     printf("Top element is: %d\n", peek());
31     pop();
32     printf("Top element after pop is: %d\n", peek());
33     return 0;

```

Test Input

stdin input (optional)

Output

Top element is: 30
Top element after pop is: 20

Pending manual review

Submitted

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INDUJEEVARA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Back

QUESTIONS

Q1 Stack 3 marks

Q2 Stack Overflow 3 marks

Q3 Stack 3 marks

Q4 Stack 3 marks

Q5 Variable Uninitialized var 3 marks

Q6 Queue 3 marks

Q7 Queue - Dequeue 3 marks

Q8 Binary Search 3 marks

Q9 Vector Queue 3 marks

Q10 Stack - Push Pop Equivalent 3 marks

10/10 answered

Q6 Queue 3 marks

Circular Queue

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```

1 #include <iostream.h>
2 #define MAX 5
3 int queue[MAX];
4 int front = -1, rear = -1;
5 void enqueue(int value) {
6     if (front == rear) {
7         printf("Queue is Full\n");
8         return;
9     }
10    queue[rear] = value;
11    rear = (rear + 1) % MAX;
12 }
13 void dequeue() {
14     if (front == rear) {
15         printf("Queue is Empty\n");
16         return;
17     }
18     front = (front + 1) % MAX;
19 }
20 int main() {
21     enqueue(10);
22     enqueue(20);
23     enqueue(30);
24     enqueue(40);
25     enqueue(50);
26     dequeue();
27     printf("Front element is: %d\n", queue[front]);
28     return 0;

```

Test Input

stdin input (optional)

Output

Front element is: 10
Front element is: 20
Front element is: 30
Front element is: 40
Front element is: 50
Front element is: 10

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TRUJENORA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)
Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Queue Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Prefix Expression
5 marks

10/10 answered

Q1 Stack
5 marks

Identify the issue in the below pseudocode.
PORG:
If top <= MAX → Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 #define MAX 5
3 int stack[MAX];
4 int top = -1;
5 void push(int val) {
6     if (top == MAX) {
7         printf("Overflow\n");
8         return;
9     }
10    stack[++top] = val;
11 }
12 int pop() {
13     if (top == -1) {
14         printf("Underflow\n");
15         return -1;
16     }
17     return stack[top--];
18 }
19 int main() {
20     push(10);
21     push(20);
22     push(30);
23     printf("Popped: %d\n", pop());
24     printf("Popped: %d\n", pop());
25     printf("Popped: %d\n", pop());
26     printf("Popped: %d\n", pop());
27     return 0;
28 }

```

Submitted

Test Input

stdin input (optional)

Output

Popped: 30
Popped: 20
Popped: 10
Underflow
Popped: -1

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SIMATS Online Programming Lab
DS PROGRAMMING

RAGUL M
TRUJENORA

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)
Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Queue Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack - Prefix Expression
5 marks

10/10 answered

Q1 Stack
5 marks

Identify the issue in the below pseudocode.
PORG:
If top <= 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 #include <stdio.h>
2 #include <stdlib.h>
3 #define MAX 5
4 int stack[MAX];
5 int top = -1;
6 void PUSH(int value) {
7     if (top == MAX) {
8         printf("Overflow\n");
9         return;
10    }
11    stack[++top] = value;
12 }
13 int POP() {
14     if (top == -1) {
15         printf("Underflow\n");
16         return -1;
17     }
18     return stack[top--];
19 }
20 int main() {
21     PUSH(10);
22     PUSH(20);
23     PUSH(30);
24     printf("Popped: %d\n", POP());
25     printf("Popped: %d\n", POP());
26     printf("Popped: %d\n", POP());
27     printf("Popped: %d\n", POP());
28     return 0;
29 }

```

Submitted

Test Input

stdin input (optional)

Output

Popped: 30
Popped: 20
Popped: 10
Underflow
Popped: -1

Pending manual review

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CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1
Stack
5 marks
- Q2
Circle Queue
5 marks
- Q3
Stack
5 marks
- Q4
Stack
5 marks
- Q5
Circular Linked List
5 marks
- Q6
Queue
5 marks
- Q7
Queue - Deque
5 marks
- Q8
Binary Search
5 marks
- Q9
Circle Queue
5 marks
- Q10
Stack - Prefix Expression
5 marks

10/10 answered

Q9 Queue - Deque

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>  
#include <stdlib.h>  
#define max 5  
typedef struct {  
    int deque[max];  
    int front;  
    int rear;  
} queue;  
void initqueue(queue *dq) {  
    dq->front = -1;  
    dq->rear = -1;  
}  
int isfull(queue *dq) {  
    return ((dq->front == 0 && dq->rear == max - 1) || (dq->front == dq->rear));  
}  
int isempty(queue *dq) {  
    return (dq->front == -1);  
}  
void insertrear(queue *dq, int data) {  
    if (isfull(dq)) {  
        printf("Overflow: queue is full!\n");  
        return;  
    }  
    if (dq->front == -1) {  
        dq->front = 0;  
        dq->rear = 0;  
    } else if (dq->front == 0) {  
        dq->front = max - 1;  
    } else {  
        dq->front = dq->front - 1;  
    }  
    dq->deque[dq->rear] = data;  
    printf("inserted %d at rear. front: %d, rear: %d\n", data, dq->front, dq->rear);  
    dq->rear = dq->rear + 1;  
}  
void insertfront(queue *dq, int data) {  
    if (isfull(dq)) {  
        printf("Overflow: queue is full!\n");  
        return;  
    }  
    if (dq->front == -1) {  
        dq->front = 0;  
        dq->rear = 0;  
    } else if (dq->rear == max - 1) {  
        dq->rear = 0;  
    } else {  
        dq->rear = dq->rear + 1;  
    }  
    dq->deque[dq->rear] = data;  
    printf("inserted %d at rear. front: %d, rear: %d\n", data, dq->front, dq->rear);  
}  
void display(queue *dq) {  
    if (isempty(dq)) {  
        printf("queue is empty!\n");  
        return;  
    }  
    printf("queue elements: ");  
    int i = dq->front;  
    while (1) {  
        printf("%d ", dq->deque[i]);  
        if (i == dq->rear) break;  
        i = (i + 1) % max;  
    }  
    printf("\n");  
}  
int main() {  
    queue dq;  
    initqueue(&dq);  
    printf("==== testing queue Operations ====\n");  
    insertrear(&dq, 10);  
    insertrear(&dq, 20);  
    insertrear(&dq, 30);  
    display(&dq);  
    insertfront(&dq, 0);  
    insertfront(&dq, 1);  
    display(&dq);  
    insertrear(&dq, 40);  
    insertrear(&dq, 50);  
    display(&dq);  
    insertfront(&dq, 60);  
    return 0;  
}
```

Test Input

state input (optional)

Output

```
==== testing queue Operations ====  
inserted 10 at rear. front: 0, rear: 0  
inserted 20 at rear. front: 0, rear: 1
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Prefix Expression
5 marks

10/10 answered

Q8 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {  
    int mid = l + (r - l) / 2;  
    if (arr[mid] == x) return mid;  
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);  
    return binarySearch(arr, mid, r, x);  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 int binarySearch(int arr[], int l, int r, int x) {  
3     if (l > r) return -1;  
4     int mid = l + (r - l) / 2;  
5     if (arr[mid] == x) return mid;  
6     if (arr[mid] > x) return binarySearch(arr, l, mid - 1, x);  
7     return binarySearch(arr, mid + 1, r, x);  
8 }  
9 int main() {  
10     int arr[] = {1, 3, 5, 7, 9, 11, 13, 15};  
11     int n = sizeof(arr) / sizeof(arr[0]);  
12     int x = 11;  
13     int result = binarySearch(arr, 0, n - 1, x);  
14     if (result != -1)  
15         printf("Element %d found at index %d\n", x, result);  
16     else  
17         printf("Element %d not found\n", x);  
18     x = 8;  
19     result = binarySearch(arr, 0, n - 1, x);  
20     if (result != -1)  
21         printf("Element %d found at index %d\n", x, result);  
22     else  
23         printf("Element %d not found\n", x);  
24     return 0;  
25 }
```

Test Input

stdin input (optional)

Output

```
Element 11 found at index 5  
Element 8 not found
```

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Submitted

CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1
Stack
5 marks
- Q2
Circular Queue
5 marks
- Q3
Stack
5 marks
- Q4
Stack
5 marks
- Q5
Unstable Linked List
5 marks
- Q6
Queue
5 marks
- Q7
Queue - Linked
5 marks
- Q8
Binary Search
5 marks
- Q9
Circular Queue
5 marks
- Q10
Stack - Prefix Expression
5 marks

10/10 answered

Q9 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic:

```
int dequeue() {  
    if (front == -1) return -1;  
    int data = queue[front];  
    front = (front + 1) % MAX;  
    return data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #define MAX 5  
3 int queue[MAX];  
4 int front = -1, rear = -1;  
5  
6 int enqueue(int value) {  
7     if ((rear + 1) % MAX == front) {  
8         return 0;  
9     }  
10    if (front == -1) {  
11        front = rear = 0;  
12    } else {  
13        rear = (rear + 1) % MAX;  
14    }  
15    queue[rear] = value;  
16    return 1;  
17 }  
18  
19 int dequeue() {  
20    if (front == -1) {  
21        return -1;  
22    }  
23  
24    int data = queue[front];  
25    if (front == rear) {  
26        front = -1;  
27        rear = -1;  
28    } else {  
29        front = (front + 1) % MAX;  
30    }  
31    return data;  
32 }  
33  
34 void printQueue() {  
35    if (front == -1) {  
36        printf("Queue is empty\n");  
37        return;  
38    }  
39    printf("Queue elements: ");  
40    int i = front;  
41    while (1) {  
42        printf("%d ", queue[i]);  
43        if (i == rear) break;  
44        i = (i + 1) % MAX;  
45    }  
46    printf("\n");  
47 }  
48  
49 int main() {  
50    printf("Test 1: dequeue from empty queue\n");  
51    printf("dequeue: %d (expected -1)\n", dequeue());  
52    printQueue();  
53    printf("\n");  
54    printf("Test 2: enqueue 10,20,30 then dequeue twice\n");  
55    enqueue(10);  
56    enqueue(20);  
57    enqueue(30);  
58    printQueue();  
59    printf("dequeue: %d (expected 10)\n", dequeue());  
60    printf("dequeue: %d (expected 20)\n", dequeue());  
61    printQueue();  
62    printf("\n");  
63    printf("Test 3: make queue have one element and dequeue it\n");  
64    printf("dequeue: %d (expected 10)\n", dequeue());  
65    printQueue();  
66    printf("dequeue: %d (expected -1)\n", dequeue());  
67    printQueue();  
68    printf("Test 4: Circular wrap-around\n");  
69    enqueue(1);  
70    enqueue(2);  
71    enqueue(3);  
72    enqueue(4);  
73    printQueue();  
74    printf("dequeue: %d (expected 1)\n", dequeue());  
75    printf("dequeue: %d (expected 2)\n", dequeue());  
76    printQueue();  
77    enqueue(5);  
78    enqueue(6);  
79    while (1) {  
80        int n = dequeue();  
81        if (n == -1) break;  
82        printf("dequeue: %d\n", n);  
83    }  
84    printQueue();  
85    return 0;  
86 }
```

Submitted

Test Input

stdin: Input (test.html)

Output

```
Test 1: dequeue from empty  
queue  
dequeue: -1 (expected -1)  
Queue is empty  
  
Test 2: enqueue 10,20,30 then  
dequeue twice
```

Pending manual review

CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack

1 marks

Q2

Circular Queue

1 marks

Q3

Stack

1 marks

Q4

Stack

1 marks

Q5

Circular Linked List

1 marks

Q6

Queue

1 marks

Q7

Queue - Deque

1 marks

Q8

Binary Search

1 marks

Q9

Circular Queue

1 marks

Q10

Stack - Postfix Expression

1 marks

10/10 answered

Q10 Stack - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {
    if (isdigit(expression[i])) push(expression[i] - '0');
    else {
        int op1 = pop();
        int op2 = pop();
        // perform operation: result = op1 + op2;
        push(result);
    }
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <ctype.h>
3 #include <string.h>
4 #define MAX 1000
5 typedef struct {
6     int a[100];
7     int top;
8 } Stack;
9 void init(Stack *s) {
10     s->top = -1;
11 }
12 int isempty(Stack *s) {
13     return s->top == -1;
14 }
15 void push(Stack *s, int x) {
16     if (s->top >= MAX - 1) return;
17     s->a[++s->top] = x;
18 }
19 int pop(Stack *s) {
20     if (isempty(s)) return 0;
21     return s->a[s->top--];
22 }
23 int evalpostfix(char *exp) {
24     Stack s;
25     init(&s);
26     int i = 0;
27     while (exp[i] != '\0') {
28         if (exp[i] == ' ' || exp[i] == '\n' || exp[i] == '\t') {
29             i++;
30             continue;
31         }
32         if (isdigit((unsigned char)exp[i])) {
33             int num = 0;
34             while (isdigit((unsigned char)exp[i])) {
35                 num = num * 10 + (exp[i] - '0');
36                 i++;
37             }
38             push(&s, num);
39             continue;
40         }
41         char op = exp[i];
42         int op1 = pop(&s);
43         int op2 = pop(&s);
44         int result = 0;
45
46         switch (op) {
47             case '+': result = op2 + op1; break;
48             case '-': result = op2 - op1; break;
49             case '*': result = op2 * op1; break;
50             case '/': result = op2 / op1; break;
51             default: result = 0; break;
52         }
53
54         push(&s, result);
55         i++;
56     }
57     return pop(&s);
58 }
59
60 int main() {
61     char exp[1000];
62     char *tests[] = {
63         "2 3 +",
64         "5 1 2 + 4 + + 3 -",
65         "10 2 /",
66         "7 3 -",
67         "13 3 + 4 +"
68     };
69
70     for (int i = 0; i < 5; i++) {
71         strcpy(exp, tests[i]);
72         printf("expression: %s\n", exp);
73         printf("result: %d\n", evalpostfix(exp));
74     }
75     return 0;
76 }
```

Submitted

Test Input:

stdin input (optional)

Output:

```
expression: 2 3 +
result: 5
expression: 5 1 2 + 4 + + 3 -
result: 14
```

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